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LSTM-Based Intention Classification System for PV Driving Direction using SSVEP and Gaze Position of Users

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Abstract: As the aging society progresses, the demand for personal vehicles (PVs) is increasing to support elderly mobility. Research has focused on improving the operability of PVs and applying autonomous driving technologies. Mixed reality (MR)-based training systems improve driving skills, but the system relies on manual operation, which raises safety concerns. Therefore a brain-computer interface (BCI) using steady-state visual evoked potentials (SSVEPs) provides an alternative for autonomous driving. Although lowering the confidence threshold improves the detection speed, however it could increases the false detections, requiring reliability improvements. This study proposes an LSTM-based deep learning system to predict a driver's steering direction using SSVEP confidence values and gaze data.

Keywords: Personal vehicle, Automatic driving, Mixed Reality, BCI, SSVEP

1. INTRODUCTION

With the aging of society, personal vehicles (PVs) are expected to become important mobility devices for the elderly. PVs can support the daily lives of the elderly by enabling them to move around at their own will, even if they have reduced the mobility. In the operation of PVs, research on improving system operability and applying autonomous driving technologies has attracted much attention.

In recent years, research on the mixed reality (MR), which overlays real and virtual spaces, has progressed, and PVs operation support and training systems have been developed [1]. Although the MR-based training systems improve the driving skills of beginners, the scheme is based on manual driving and cannot guarantee the user's driving safety. Therefore, a new scheme for selecting driving routes without precise manual operation is required.

A new approach to this problem is using a brain-computer interface (BCI). A BCI is an interface that controls computers and devices from information emitted using the human brain, such as brain waves. Among them, a BCI using visual evoked potentials (VEPs) can detect the electrical signals evoked by visual stimulation, as the electroencephalogram (EEG).[2]. In particular, the small and inexpensive devices have been developed to detect the steady-state visual evoked potentials (SSVEPs), and used for studies such as arm control and electric wheelchair operations[3][4]. The authors have developed an autonomous driving system using the SSVEPs and the MR[8]. The Intention detection using the SSVEPs has a relatively high signal-to-noise ratio compared to other BCI methods, but the detection time is long. The SSVEPs can be detected faster by lowering the threshold of the confidence value for the steady-state determination. However, as the reliability of Intention detection decreases and the number of false detections increases, it is necessary to compensate for the reliability.

There is a method to estimate the driving direction using deep learning. Previous studies have used the CNN [5] and the Long Short-Term Memory (LSTM) [6] to classify drowsiness while driving, classified into eight stages using multiple information such as the electroencephalogram (EEG) and the heart rate variability (HRV).[7]. From these studies, it is expected that driving direction can be accurately classified by using not only SSVEPs but also various information of the driver to detect the intention for autonomous driving.

In this paper, the accuracy of driving direction classification is improved by using deep learning to determine gaze information in addition to confidence values, which are the attention level of SSVEPs, to detect driving intention, addressing the limitation of our prior system [8], in which lowering the threshold of the confidence value achieve faster detection increased false detections. The LSTM is used for deep learning, aiming to complement reliability even in environments with pedestrians and build a model that effectively adapts to false detection. To collect data for training, we measured data in an environment with multiple pedestrians walking. Finally, the collected data is used to train an LSTM model and evaluate its performance.

2. EXPERIMENTAL SYSTEM CONFIGURATION

Fig. 1 shows the hardware configuration of the automatic driving support system using SSVEPs. The left side of the figure is the automatic operation block. Fig. 2 shows the PV, the WHILL model CR. The entire system is controlled by ROS, a robotic software platform. The Navigation Stack, a framework of ROS for automatic driving systems, is used for automatic driving. The Navigation Stack consists of amcl, which estimates the cur-

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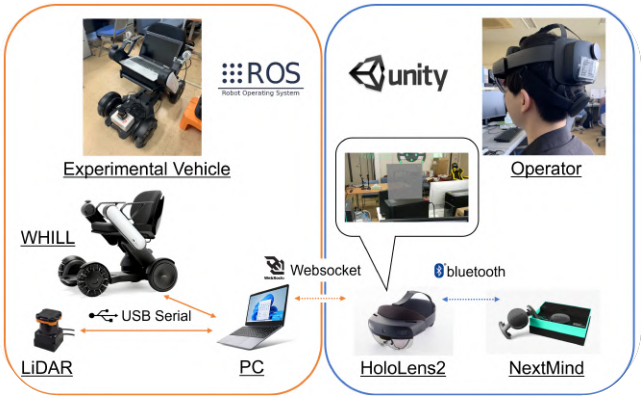
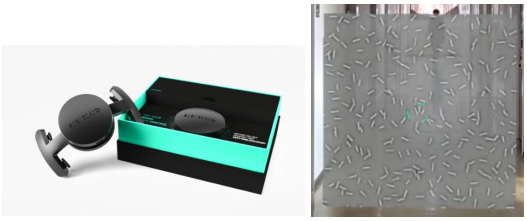


Fig. 1.Hardware configuration of automatic drive system with BCI.



Fig. 2.Driving scene of automatic driving PV with BCI.



(a) NextMind. (b) NeuroTag.

Fig. 3.NextMind and NeuroTag.

rent location, map_server, which manages the map, and move_base, which generates speed command values.

The path planning algorithm used is navfn for the Global Planner and the Dynamic Window Approach (DWA) for the Local Planner. The navfn is an algorithm that uses Dijkstra's method to find the least expensive path. The DWA selects the feasible candidate local paths from the current robot's velocity and generates the local path that avoids obstacles using an evaluation function.

The right side of Fig. 1 shows the devices that interface with the user, where the MR device is the HoloLens2 and the BCI device is the NextMind. The measurement target is projected onto the HMD used by the user. The NextMind detects the user's SSVEP, which sequentially updates the navigation goals for automatic driving.

An SDK for application development is provided when using NextMind, and NextMind applications are developed in the MR space using Unity. Fig. 3(a) shows the NextMind and Fig. 3(b) shows the NeuroTag. NeuroTag can detect the SSVEP of a user who is gazing at a small white line that flickers at a certain frequency.

3. AUTOMATIC DRIVING SYSTEMS USING BCI

3.1 Overview of traveling intention detection

The automatic driving system, which sequentially updates the navigation goals by intention detection using the BCI, is summarized. The measurement objects are placed on the HMD at each intersection along the path. To determine the intended travel direction, measurement objects are placed at each approach of the intersection. The subject gazes at the measured object in the desired direction and moves to the direction where the confidence value reaches 1. Fig. 5 shows an example of the measurement objects at an intersection.

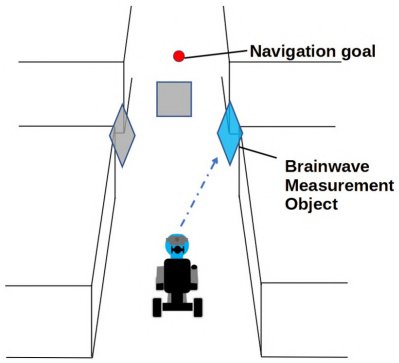


Fig. 4.Select the direction using BCI device.

The measurement object appears when the PV approaches 3m before the intersection. The range of 3m from the position where the measured object appeared at the intersection is designated as the intention detection area, and the transition of the confidence value is recorded within this area to estimate the traveling intention.

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(a) Closer to 3m.



(b) Within 3m.

Fig. 5.Virtual objects indicating the driving direction.

3.2 Measurement of confidence value

First, we describe the basic experiments with NextMind. Fig. 6(a) shows the experiment image, and Fig. 6(b) shows the measurement screen. A 0.1m square measurement object is placed 0.5m away from the subject, and the time until the confidence value reaches 1 is measured 10times.



(a) Experiment image. (b) Measurement screen.

Fig. 6.Evaluation of NextMind.

The SSVEP detection times were compared between a group familiar with NextMind and a group of novices. Table 1 shows the SSVEP detection times for the experts and Table 2 shows the times for the novices. Table 1 and Table 2 show that the average detection time increases by more than 3 seconds for beginners and has a large variation. It is difficult for novice users to detect SSVEP stably.

Table 1.Detection time (Expert).

	Min	Max	Mean	SD
Time	1.58s	2.64s	2.03s	0.94

Table 2.: Detection time (Beginner)

	Min	Max	Mean	SD
Time	2.73s	8.13s	5.31s	2.06

Fig. 7 shows the transition of the confidence values of the experts. The confidence value is characterized by a rapid increase to 1 from around 0.20 to 0.30 after gazing at the measured object. The threshold for the confidence value is 1 by default, but even experts can remain around 0.20–0.3 if their attention is not properly directed. According to the questionnaire, subjects were careful during the suspension period, with around 0.20–0.30. The results of the experiment showed that the gaze point of the HMD was shifted due to the vibration of the road surface, which may be the reason why the gaze did not continue for a certain period of time.

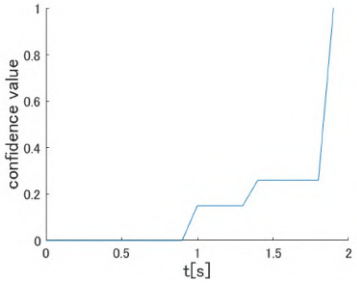


Fig. 7.Example of confidence value.

Gazing at a virtual object for a long time is painful. Lowering the threshold to around 0.20 can greatly reduce the detection time, especially for beginners. However, lowering the threshold value to speed up the detection time causes a decrease in detection accuracy and an increase in false detections. For example, it is known that when multiple pedestrians are walking, the user becomes distracted by them and the confidence value is maintained at a low value. Therefore, when the threshold of the confidence value is lowered, classification of the user's driving direction is attempted using the movement of the user's gaze point within the detection area.



Fig. 8.Pedestrians walking on the road.

4. OVERVIEW OF LONG SHORT-TERM MEMORY

4.1 Configuration of LSTM

The LSTM is an algorithm that improves on the RNN to consider time series. Because it can handle time-series data, it is used for natural language processing and data estimation. The difference between LSTM and RNN is the memory cell. The RNNs have no memory cells, so the information computed in one layer of the neural network is directly transferred to the next layer. On the other hand, the LSTM has a memory cell as shown in Fig. 9,

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which allows machine learning while keeping some of the input information. The RNNs without memory cells have a problem that machine learning is not efficient because the feature called gradient is lost during the computation process in many layers. The LSTM solves the gradient loss problem of RNNs by storing the necessary data in memory cells.

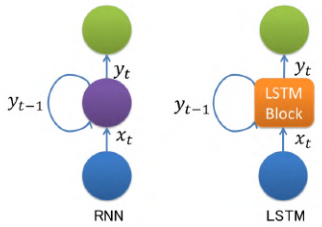


Fig. 9.Difference between RNN and LSTM.

Fig. 10 shows the block structure called a memory cell of LSTM. The LSTM uses a memory cell (cell state) and three gates (forget gate, input gate, and output gate) to control information. Each gate outputs coefficients to adjust the degree to which it forgets, ignores, or retains the memory cell, and learns how much information to put in and out of the cell in each frame.

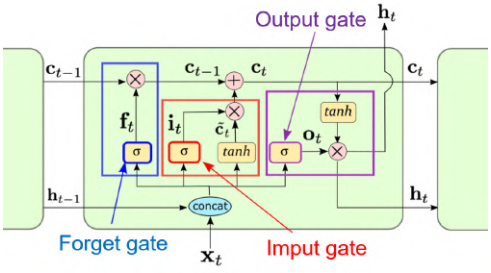


Fig. 10.LSTM block.

4.2 Evaluation metrics for LSTM

The following metrics are used to evaluate the performance of the model.

4.2.1 Categorical crossentropy loss

The categorical crossentropy was used as the loss function of the model. The metric measures the error between the model's output probability distribution and the actual label distribution. The loss value is defined by the following equation.

$$\text{Loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log(p_{ij}) \tag{1}$$

Where N is the total number of data, C is the number of classes, y_{ij} is the ground truth label of the i-th sample in one-hot representation, and p_{ij} is the predicted probability output by the model.

4.2.2 Accuracy

Accuracy is the metric that shows the percentage of correct class predictions in the total data. This metric is an overall evaluation metric that is appropriate when there is no class imbalance in the data. The definition is as follows.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{2}$$

4.2.3 Precision

Precision is the metric that the percentage of data that the model predicts to be in the correct class among the data that are actually in the correct class. Here, the TP is the true positive, and the FP is the false positive. The definition is as follows.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3}$$

4.2.4 Recall

Recall is a metric that the percentage of data that are actually in the correct class that the model predicts to be in the correct class. The definition is as follows. The FN is the false negative.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4}$$

4.2.5 F1-Score

F1-Score is the harmonic mean of the Precision and the Recall, and is a metric that evaluates the balance between them. The definition is as follows.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

The closer the F1-Score is to 1, the higher both Precision and Recall are at the same time, which means that the learning results are the most efficient and balanced, and therefore more accurate and reliable.

The Macro Average, which is the average of the Precision, Recall, and F1-Score calculated for each class, and the Weighted Average, which is the average weighted according to the number of data in a class, are also used for evaluation.

4.2.6 Confusion matrix

Confusion matrix is used to analyze the prediction results for each class. The matrix is a visualization of the performance of a model or classification algorithm in a machine learning process in matrix form, and is used as a metric to measure the performance of a binary classification machine learning model. The confusion matrices aid in predictive analysis and are used to evaluate the quality of predictive models, and are used in supervised learning. The rows of the matrix represent instances of the classes predicted by the model, and the columns represent instances of the actual classes. This matrix is as follows.

The confusion matrix allows us to visually understand the misclassification of each class.

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Table 3.Example of confusion matrix.

	Predicted Class 0	**Predicted Class 1**	**Predicted Class 2**
Actual Class 0	True Positive (TP)	False Positive (FP)	False Positive (FP)
Actual Class 1	False Negative (FN)	True Positive (TP)	False Negative (FN)
Actual Class 2	False Negative (FN)	False Negative (FN)	True Positive (TP)

5. DETECTION OF TRAVELING INTENTION FOR AUTOMATIC DRIVING

5.1 LSTM configuration

The configuration of the neural network that classifies the user's traveling direction is shown in Fig. 11. The inputs are C_l, C_c, and C_r, which are the confidence values for each direction, and D_l, D_c, and D_r, which are the reciprocals of the distance between the user's gaze point and the center of each measurement object. The outputs of the neural network are left, center, and right, which represent the traveling direction.

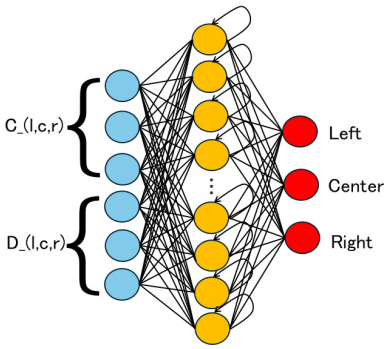


Fig. 11.Design of neural networks.

Fig. 12 shows the behavior of the neural network. The time when $C_l = 0.20$ is set to 0s. When any of the confidence values exceeds the threshold value, $C_{l,c,r}$ and $D_{l,c,r}$ in the period $[T_{set}, 0]$ are input to the NN, and the learned LSTM determines the driving direction.

Table 4 shows the learning parameters of LSTM.

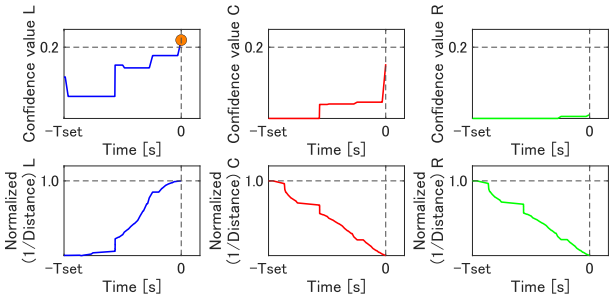


Fig. 12.Image of confidence value.

5.2 Learning results of LSTM

This study considered the following three cases for learning:

Table 4.Hyper parameter settings.

Hyper parameter	**Value**
Batch size	8
Epochs	500
Hidden layer	50
Window size	5
Stride	1
Learning rate	0.001
Dropout rate	0.2

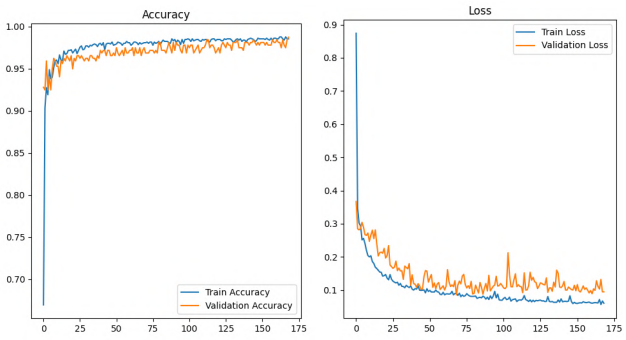
1. Using data until 1.25 seconds before detection, and confidence values as input for training.
2. Using data until 1.25 seconds before detection, the confidence value and the distance between the gazing point and the center of each object are used as input for training.
3. Using data until 2.5 seconds before detection, the confidence value and the distance between the gazing point and the center of each object are used as input for training.

There were no experiments for durations shorter than 1.25 seconds because the number of SSVEP data would be small.

Table 5 and Fig. 13 and 14 show the learning results for case 2. The training data were prepared for 44 data sets each for left, straight ahead, and right. The verification data were prepared for 12 data sets each for left, straight ahead, and right. It means that 1192 of training data and 328 of validation data were prepared, with the period from the end of detection to 1.25 second as one unit. The data used for training is resized to ensure the same number of samples.

Table 5.Confusion matrix of condition 2.

Label	**Precision**	**Recall**	**F1-Score**	**Data number**
Right	0.99	0.97	0.98	108
Left	0.97	1.00	0.99	108
Center	0.99	0.98	0.99	108
Macro Ave.	0.98	0.98	0.98	324
Weighted Ave.	0.98	0.98	0.98	324



(a) Accuracy. (b) Loss.

Fig. 13.Accuracy and loss (condition 2).

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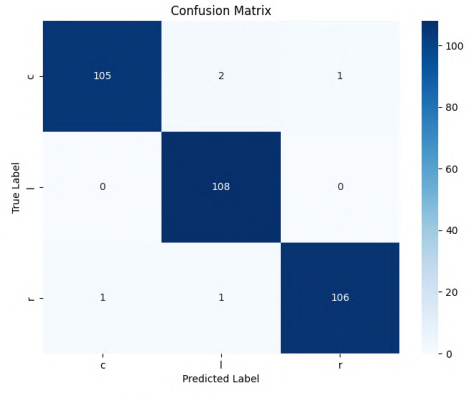


Fig. 14.Classification result (condition 2).

Table 5 shows that the F1-Score is 0.95 or higher for all classes, and the F1-Score of the Macro Average is also 0.95, showing that the classification is highly accurate. Fig. 13 shows that the accuracy in the validation data corresponds to the accuracy in the training data. The loss value for the training data tended toward 0, while the loss value for the validation data reached an equilibrium point below 0.1. The learning process was stopped at 170 epochs with early stopping to avoid overtraining.

5.3 Performance evaluation

In the performance evaluation, test data is provided to the models created under each condition to verify the generality of the models. A total of 4 of test data was prepared for the experiment: 1 of left, straight and right test data and 1 of false positive data. The results are shown in Table 6.

Table 6.Performance evaluation.

Model	Correct rate (%)
Condition 1	81.25
Condition 2	100.0
Condition 3	98.14

If the confidence value threshold is 1, the detection accuracy is 100%. When the threshold value is 0.2, the detection is also fast, but Table 6 shows that there is a large difference in accuracy for each case. Case 1, which uses only confidence values, has a low accuracy rate of 81.25%, resulting in misclassifications during the detection process. Case 2, which uses confidence values and gaze data, shows 100% accuracy, showing that LSTM has generalization performance. Case 3 uses the same information as case 2, but with twice the number of time steps. In this case, the accuracy is 98.14%, which is lower than in case 2. Although the data are omitted, the equilibrium point of the loss values is slightly larger than in case 2, resulting in lower generalization performance. The loss is also degraded, and its amplitude is increased. From the perspective of model generalization, these results show that the performance is generally improved when multiple input features are used. The data length may also be related to the traffic volume and the PV speed, which should be considered.

6. CONCLUSION

In this paper, the driving direction was classified by using deep learning to determine the confidence value of the SSVEP and eye information for detecting the user's driving intention. In the experiment, the LSTM model was trained in environments without pedestrians and in environments with multiple pedestrian intersections. The classification performance of driving direction in automatic driving was greatly improved by adding the distance between the gaze point and the center of each object to the confidence value as an input for training.

From the perspective of model generalization, it was found that the performance is greatly improved when multiple input features are used. So, it is expected that advanced classification can be achieved by using pedestrian information, such as the pedestrian location and traveling direction, in the learning process.

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